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MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

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SUMMARY IN SPANISH

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This research addresses the challenge of continual learning for object classification using Gaussian Modeling. It focuses on integrating foundation models as pretrained feature extractors with independent Gaussian models to characterize object classes. The main objective of this work is to develop and validate a continual learning framework for object classification, using pre-trained foundation models as features extractors and independent Gaussian models, with the purpose of enabling the incremental incorporation of new classes while preventing catastrophic forgetting of previously learned class. To achieve this, the study evaluates the effectiveness of various Gaussian representations and characterizes the used methods. The proposed approach seeks to combine the independence of Gaussian models with the generalization capacity of foundation models as feature extractors, providing a scalable and efficient solution for dynamic environments where new object classes continuously emerge.

Abstract

ABSTRACT IN SPANISH

Abstract

Traditional deep learning architectures are predominantly monolithic, often failing to adapt when exposed to non-stationary data distributions. Incorporating new classes typically necessitates exhaustive retraining or fine-tuning on the entire dataset to prevent catastrophic forgetting, a constraint that limits their deployment in dynamic, real-world systems. To overcome these challenges, this work introduces a modular Continual Learning (CL) framework that decouples feature extraction from classification. By leveraging a frozen, pre-trained feature extractor paired with independent Gaussian models, we create distinct probabilistic representations for each class. This architectural decoupling allows for the seamless, incremental addition of new categories without interfering with previously acquired knowledge. Extensive evaluation on rigorous benchmarks —including Split CIFAR-100, Split ImageNet-R, and Caltech-256— demonstrates that our method achieves performance competitive with current state-of-the-art CL approaches. Furthermore, the results highlight that the proposed strategy is significantly more scalable, computationally lighter, and simpler to implement than existing replay-based or regularization-intensive methods, offering a robust solution for efficient continual learning.

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Chapter 1

Introduction

1.1 Motivation

State-of-the-art (SOTA) object classification methods utilize deep learning models that are either trained end-to-end on the target application or initially trained in one application and then fine-tuned for a different target application. In either case, they are monolithic models: the same single model is normally trained on a vast dataset and used for feature extraction and classification. These models are highly effective when the set of object classes to be recognized is fixed and known at the training/fine-tuning time. However, when new object classes need to be incorporated, the model must be retrained or fine-tuned. This training requires significant computational resources and access to the entire training dataset and, in some cases, leads to catastrophic forgetting [4].

One way to integrate new classes while avoiding catastrophic forgetting, is to approach the problem using continual or class-incremental object classification, where an algorithm must learn to distinguish between a growing number of object classes [5]. In the context of continual learning, a task refers to a distinct portion of the training experience, usually presented sequentially to the model. Instead of having access to the entire dataset at once, as in standard supervised learning, the model receives data in sequential chunks [6]. Each task generally correspond to a set of new classes to recognize. The main challenge is that when the model is trained task by task, it tends to overwrite previously learned knowledge. Given that current monolithic models are not capable of handling this type of application, an alternative paradigm is required.

This limitation has motivated SOTA methods for object classification to focus on balancing stability and plasticity. Stability refers to a model's ability to preserve previously learned knowledge, while plasticity describes its capacity to adapt and incorporate new tasks [7]. However, traditional approaches rarely optimize both aspects simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones, and by the need to continually adjust the network's structure to match the increasing number of

object classes.

One solution to this problem is to divide the monolithic models into two distinct components that must be trained separately, at different times: one responsible for feature extraction and the other for object classification. The feature extraction model must be trained when the recognition system is initialized/created and the obtained features must be general enough to characterize all possible classes of objects to be seen (not just the ones known at training time). The object classification model should be easily updated or trained as new object classes are detected. Ideally, it should be able to incorporate new object classes without modifying the parts of the model responsible for recognizing the already known object classes. This mirrors the functionality of the human visual system, where the structures responsible for the extraction of characteristics (simple and complex cells in cortical areas V1 and V2) undergo adaptation during the initial postnatal weeks/months [8]. However, as object recognition occurs in high-level cognitive areas, and acquiring knowledge of a new object class can result in a slight reduction in the ability to recognize similar classes of objects over time, the ability to learn new categories of objects persists throughout life.

We are inspired by this idea and propose a continual learning object recognition system based on two components. The first component is a feature extraction model built using a pre-trained foundation model. The second component is a statistical classifier built using various Gaussian models. Each model characterizes a specific object class independently of the others. Since these Gaussian models can be trained or built at different times, a new Gaussian model is created to characterize a new object class when it needs to be incorporated. For classification, the class whose Gaussian model yields the highest likelihood of the input features is selected. This approach is analogous to a Bayes classifier, in which new classes are incorporated over time.

Gaussian models offer two key advantages for characterizing object classes: first, each object class model operates independently from the others, facilitating the straightforward incorporation of new classes into the model. Second, Gaussian models provide a compact representation of object classes, requiring few samples to construct a model that can be refined as new samples become available.

1.2 Hypothesis

We hypothesize that the combination of independent Gaussian models and a pretrained feature extractor provides a robust framework for continual object classification. By enabling the addition of new classes without full-model retraining, this design eliminates catastrophic forgetting. The inherent simplicity of this approach is a key advantage, yielding performance that rivals state-of-the-art methods while maintaining lower computational overhead.

1.3 General Objectives

The main objective of this work is to develop and validate a continual learning framework for object classification, using pre-trained foundation models as features extractors and independent Gaussian models, with the purpose of enabling the incremental incorporation of new classes while preventing catastrophic forgetting of previously learned class.

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- To assess the Gaussianity of feature distributions within the latent space to validate the underlying distributional assumptions.
- To characterize the efficacy of independent Gaussian distributions as class representations within a continual learning framework.
- To quantify the impact of various covariance matrix approximation methods on model stability and classification accuracy.
- To evaluate the generalization capabilities of diverse feature extractors across multiple dataset configurations and incremental learning scenario.

1.5 Thesis Structure

Chapter 2

Theoretical Framework

This chapter establishes the conceptual and mathematical foundations required to contextualize this research, transitioning from broad paradigms in Artificial Intelligence to the specific statistical mechanics and adaptive learning strategies underpinning our methodology. We begin by reviewing the fundamental principles of Machine Learning, specifically within the realms of image processing and object classification, before providing a rigorous mathematical treatment of Bayesian Inference, Mixture of Gaussians (MoG), and Principal Component Analysis (PCA). Furthermore, we detail advanced covariance estimation methods such as OAS and Ledoit-Wolf, which are critical for stability in high-dimensional feature spaces. Finally, we address the challenge of "catastrophic forgetting" by introducing the Continual Learning (CL) paradigm, describing the datasets utilized and evaluating the state-of-the-art mitigation strategies that provide a performance baseline for the proposed work.

2.1 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a broad field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and language understanding [9]. Within this domain, Machine Learning (ML) focuses on developing algorithms that allow computers to learn patterns from data without explicit programming. ML systems are generally categorized into three paradigms: Supervised Learning, Unsupervised Learning, and Reinforcement Learning [10].

First, in **Supervised Learning**, the model learns a mapping from input variables $\mathbf{x}$ to output variables y based on a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$. The objective is to minimize a loss function that penalizes the discrepancy between the predicted output and the ground truth. This paradigm is typically divided into Classification, where the output is a discrete category, and Regression, where the output is a continuous value.

Second, **Unsupervised Learning** deals with unlabeled data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$. Instead of predicting a target, the goal is to discover the underlying structure or distribution of the data. Common tasks include Clustering, which groups similar data points (e.g., K-Means or

Gaussian Mixture Models), and Dimensionality Reduction, which projects high-dimensional data into a lower-dimensional space while preserving essential information (e.g., PCA or t-SNE).

Finally, **Reinforcement Learning** is based on an agent interacting with an environment to maximize a cumulative reward. Unlike supervised learning, the agent is not told which actions to take but must discover them through trial and error. This process is often modeled as a Markov Decision Process (MDP) [**?**], where the agent transitions between states based on actions, receiving feedback in the form of numerical rewards. The focus is on finding an optimal policy that balances exploration (trying new actions) and exploitation (using known high-reward actions).

2.1.1 Deep Learning

In traditional Machine Learning, the feature extraction step is a manual process where a human expert must identify and define the most relevant characteristics (such as the shape or edges of the car) before passing them to a classification algorithm. In contrast, as shown in figure 2.1, Deep Learning utilizes neural networks to automate this process; the model performs both feature extraction and classification simultaneously, learning to identify complex patterns directly from the input data without human intervention.

Structurally, this shift allows Deep Learning to handle much more complex, high-dimensional data, though it typically requires significantly larger datasets and more computational power to be effective. While traditional Machine Learning depends on the quality of hand-crafted features, Deep Learning architectures—like multi-layered neural networks—are designed to build hierarchical representations of data, progressively moving from low-level pixels to high-level object concepts.

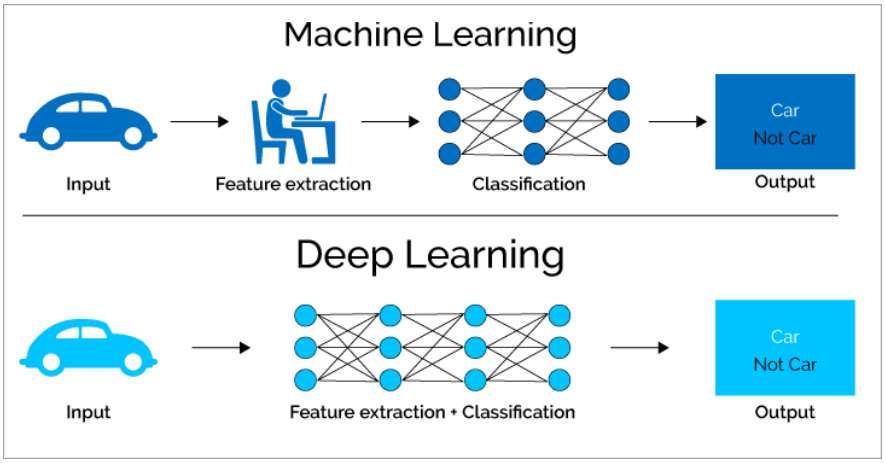


Figure 2.1: Machine Learning and Deep Learning Strategies. Image taken from [1].

2.1.2 Computer Vision and Vision Transformers

Computer Vision is the interdisciplinary subfield of AI that enables computers to gain high-level understanding from digital images or videos, effectively automating tasks performed by the human visual system [11]. While Convolutional Neural Networks (CNNs) [12] historically dominated this field, the Transformer architecture—originally developed for Natural Language Processing—has revolutionized visual recognition through the self-attention mechanism [13].

Vision Transformers (ViT) adapt this architecture by treating an image as a sequence of discrete tokens. As illustrated in Figure 2.2, the process follows a specific pipeline:

- **Patch Partitioning:** The input image is decomposed into fixed-size patches.
- **Linear Projection:** These patches are flattened and mapped to an embedding space.
- **Positional Encoding:** Since Transformers lack inherent knowledge of spatial structure, learnable position embeddings are added to the patch embeddings.
- **Transformer Encoder:** The resulting sequence is processed through multiple layers of multi-head self-attention and MLP blocks to capture global dependencies.

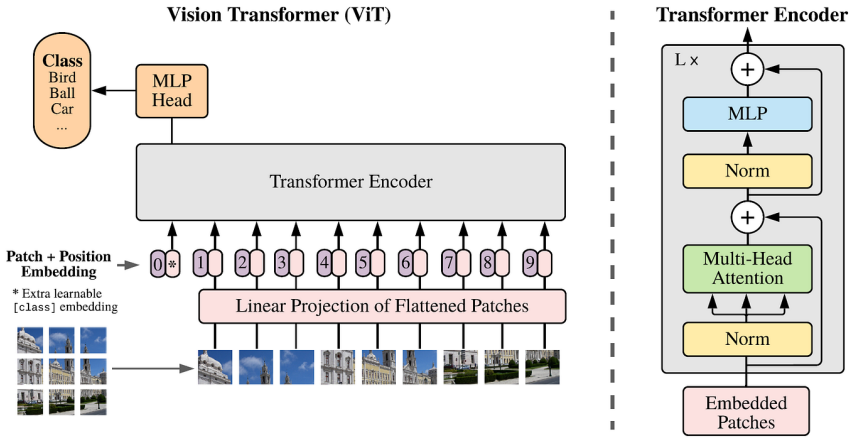


Figure 2.2: Vision Transformer (ViT) architecture overview. The model processes an image as a sequence of patches through a standard Transformer encoder. Image taken from [2].

Object Classification and Representative Power

The primary objective of the original ViT was *Object Classification*: the fundamental task of assigning a single categorical label to an entire input image. This task serves as the primary benchmark for evaluating the representative power of visual models [14]. By processing the global sequence of patches, ViTs have demonstrated an ability to meet or exceed CNN performance, particularly when pre-trained on the large-scale datasets described below.

2.1.3 Image Datasets and Benchmarks

To rigorously test the robustness and generalization of architectures like ViT, this work utilizes several standardized datasets that vary in scale, stylistic diversity, and granularity.

The ImageNet Family

ImageNet [15] is a cornerstone of computer vision, providing over 1.2 million images across 1,000 classes. Its hierarchical structure, based on WordNet, makes it the standard for large-scale classification. To further push the boundaries of model performance, two variants are often employed:

- **ImageNet-21K (and ImageNet-21k-P):** An expansive version containing 14 million images across 21,000 classes [16]. The "P" version is a cleaned subset designed to make pre-training for models like ViT more accessible.
- **ImageNet-R (Renditions):** A dataset of 30,000 images covering 200 classes in non-photographic styles (e.g., art, sketches). It tests a model's robustness to out-of-distribution data [17].

Fine-Grained and Diverse Benchmarks

Beyond large-scale pre-training, fine-grained classification is evaluated using:

- **CIFAR-100:** Consists of 60,000 low-resolution (32×32) images [18]. It is unique for its two-level hierarchy, where 100 "fine" classes are nested within 20 "coarse" superclasses, allowing for the evaluation of both specific and broad categorization.
- **Caltech-256:** Provides 30,607 images across 256 categories [19]. It offers greater category diversity and a higher minimum number of images per class than its predecessors, serving as a robust benchmark for generalization.

Self-Supervised Advancements: DINOv2

While the aforementioned datasets provide explicit labels for supervision, recent advancements have led to foundation models such as **DINOv2** [20]. Unlike earlier iterations, DINOv2 utilizes a discriminative self-supervised approach that combines contrastive learning with image-level and patch-level objectives. By training on massive, curated datasets without human labels, DINOv2 produces multipurpose visual features that exhibit strong "out-of-the-box" properties, often surpassing models trained with explicit supervision on tasks like semantic segmentation [20].

2.2 Foundations of Probabilistic Modeling

The convergence of statistical learning theory and high-dimensional data analysis has necessitated the development of robust mathematical frameworks to handle the dual challenges of uncertainty quantification and dimensionality reduction. As modern neural architectures, such as the DINOv2 Vision Transformer, extract increasingly complex representations from datasets like ImageNet-R, the subsequent classification and density estimation layers must be grounded in rigorous probabilistic foundations.

2.2.1 Likelihood-Based Classification

Likelihood inference provides a coherent framework for updating the probability of a hypothesis as more evidence or information becomes available. In contrast to frequentist approaches that treat parameters as fixed but unknown constants, the Bayesian paradigm treats parameters as random variables characterized by probability distributions.

A generative modeling framework can be adopted and class-conditional Gaussian feature distributions can be assumed:

$$p(\mathbf{z} \mid y = c) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{2.1}$$

where $\mathbf{z} \in \mathbb{R}^d$ represents the feature vector, $\boldsymbol{\mu}_c$ is the class mean vector, and $\boldsymbol{\Sigma}_c \succ 0$ is the covariance matrix.

Classification can be performed via maximum a posteriori (MAP) inference:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\mathbf{z} \mid y = c) + \log \pi_c \right], \tag{2.2}$$

where $\pi_c = p(y = c)$ denotes the class prior.

Using the Gaussian density, the decision rule is equivalent to

$$\hat{y} = \arg\min_{c} \left[(\mathbf{z} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| - 2 \log \pi_c \right]. \tag{2.3}$$

This yields a quadratic decision function in the feature space.

Mixture of Gaussians (MoG)

A Mixture of Gaussians models a complex probability density as a convex combination of K distinct Gaussian components. This approach is fundamental for clustering and density estimation in multi-modal distributions. Therefore, to capture multimodal feature distributions, the class-conditional model can be extended to a mixture:

$$p(\mathbf{z} \mid y = c) = \sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}), \tag{2.4}$$

where $\alpha_{c,k} \geq 0$ and $\sum_k \alpha_{c,k} = 1$.

The MAP decision rule becomes

$$\hat{y} = \arg\max_c \left[\log \left(\sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}) \right) + \log \pi_c \right]. \tag{2.5}$$

2.2.2 Advanced Covariance Estimation in High Dimensions

In many modern applications, such as image feature analysis using DINOv2, the dimensionality d (often in the hundreds or thousands) often exceeds the sample size n. In this "large d, small n" regime, the empirical sample covariance matrix (SCM) S is a poor estimator: its eigenvalues are biased, and the matrix is singular if $n < d$. This singularity prevents the inversion of the matrix required for the quadratic form in classification and density estimation.

Regularization and Numerical Stability

To stabilize the inversion of the covariance matrix $\boldsymbol{\Sigma}_c$ for each class, regularization techniques are employed. A primary method is diagonal loading, which adds a small constant ε (e.g., 10^{-6}) to the diagonal elements:

$$\boldsymbol{\Sigma}_{reg} = \boldsymbol{\Sigma}_c + \varepsilon I \tag{2.6}$$

This adjustment ensures that the matrix remains strictly positive definite. However, for higher numerical precision in calculating the inverse (precision matrix) and the log-determinant, we utilize Spectral Filtering via eigendecomposition.

Spectral Filtering and Precision Matrix Inversion

For each class, the system computes the eigenvectors P and eigenvalues λ such that $\boldsymbol{\Sigma}_{reg} = P\Lambda P^T$. To ensure stability against near-zero eigenvalues that survive initial regularization, we apply a floor: $\hat{\lambda} = \max(\lambda, \varepsilon)$. The precision matrix (inverse covariance) is then reconstructed as:

$$\boldsymbol{\Sigma}_c^{-1} = P \text{diag}\left(\frac{1}{\hat{\lambda}_1}, \ldots, \frac{1}{\hat{\lambda}_d} \right) P^T \tag{2.7}$$

The log-determinant, required for the Gaussian log-likelihood, is computed directly from these filtered eigenvalues to avoid underflow:

$$\ln |\mathbf{\Sigma}_c| = \sum_{j=1}^{d} \ln(\hat{\lambda}_j) \tag{2.8}$$

Finally, the normalization constant $\ln A_c$ (representing the constant term of the multivariate Gaussian log-density) is stored:

$$\ln A_c = -\frac{1}{2} \left(d \ln(2\pi) + \ln |\mathbf{\Sigma}_c| \right) \tag{2.9}$$

This approach eliminates "noise floor" issues that would otherwise cause the matrix to become singular or produce infinite values during log-likelihood estimation.

Linear Shrinkage Estimators

Linear shrinkage methods propose a compromise between the SCM S and a highly structured target matrix F, such as the identity matrix or a diagonal matrix of sample variances.

Ledoit-Wolf Estimator The Ledoit-Wolf estimator [**?**] determines an optimal shrinkage intensity ρ to minimize the expected Frobenius norm between the estimate and the true covariance matrix $\mathbf{\Sigma}$. It is defined as:

$$\hat{\mathbf{\Sigma}}_{LW} = (1 - \rho)S + \rho F \tag{2.10}$$

Ledoit-Wolf is computationally efficient and provides an asymptotically optimal estimate without assuming a specific data distribution.

Oracle Approximating Shrinkage (OAS) The OAS estimator [**?**] refines the Ledoit-Wolf approach by assuming the data follows a Gaussian distribution. This assumption allows for a more precise estimation of the shrinkage intensity, particularly for small sample sizes. The OAS shrinkage parameter ρ_{OAS} is given by:

$$\rho_{OAS} = \min \left(1, \frac{(1 - \frac{2}{d})\text{tr}(S^2) + \text{tr}^2(S)}{(n + 1 - \frac{2}{d})(\text{tr}(S^2) - \frac{\text{tr}^2(S)}{d})} \right) \tag{2.11}$$

OAS is specifically designed to minimize Mean Squared Error (MSE) under Gaussianity and often outperforms standard shrinkage when the data distribution is approximately normal.

2.2.3 Principal Component Analysis (PCA)

PCA is a linear transformation technique that identifies the directions (principal components) along which the data exhibits the highest variance. It serves as an essential tool for noise reduction and visualization in high-dimensional feature spaces.

Variance Maximization and Lagrange Multipliers

Given a centered data matrix $X \in \mathbb{R}^{n \times d}$, the goal is to find a unit vector $\mathbf{w}$ that maximizes the variance of the projected data $X\mathbf{w}$. The variance is:

$$\text{Var}(X\mathbf{w}) = \frac{1}{n}(X\mathbf{w})^T(X\mathbf{w}) = \mathbf{w}^T \left(\frac{X^T X}{n} \right) \mathbf{w} = \mathbf{w}^T S \mathbf{w} \tag{2.12}$$

where S is the sample covariance matrix. To prevent $\mathbf{w}$ from growing indefinitely, we impose the constraint $\mathbf{w}^T\mathbf{w} = 1$. Using the method of Lagrange multipliers:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T S \mathbf{w} - \lambda(\mathbf{w}^T\mathbf{w} - 1) \tag{2.13}$$

Differentiating with respect to $\mathbf{w}$ and setting to zero:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2S\mathbf{w} - 2\lambda\mathbf{w} = 0 \implies S\mathbf{w} = \lambda\mathbf{w} \tag{2.14}$$

This demonstrates that $\mathbf{w}$ must be an eigenvector of the covariance matrix S, and λ is its corresponding eigenvalue. Since $\mathbf{w}^T S \mathbf{w} = \mathbf{w}^T \lambda \mathbf{w} = \lambda$, maximizing the variance is equivalent to selecting the eigenvector associated with the largest eigenvalue.

Sequential Orthogonality

Subsequent principal components are found by maximizing variance subject to the constraint that they are orthogonal to all previous components. Inductively, the k-th principal component corresponds to the k-th largest eigenvalue. This process decomposes the total variance into uncorrelated linear combinations of the original variables.

2.3 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisition of knowledge from sequences of tasks or data streams over time. As illustrated in Figure 2.3, a **task** in this context is defined as a specific learning experience or dataset –such as Task

A, B, or C. Each task is potentially characterized by unique data distributions or objectives, as symbolized by the varying shapes (e.g., circles vs. hearts) above the task labels.

Unlike conventional training approaches that rely on a fixed dataset and a stationary environment, CL emphasizes the arrival of these tasks at distinct intervals along the **time** axis. This sequential arrival requires the model to be inherently non-stationary, adapting its internal representations as environments evolve. The primary challenge in this paradigm lies in the ongoing integration of new information from a current task (e.g., Task B) while simultaneously preserving the knowledge previously acquired from Task A.

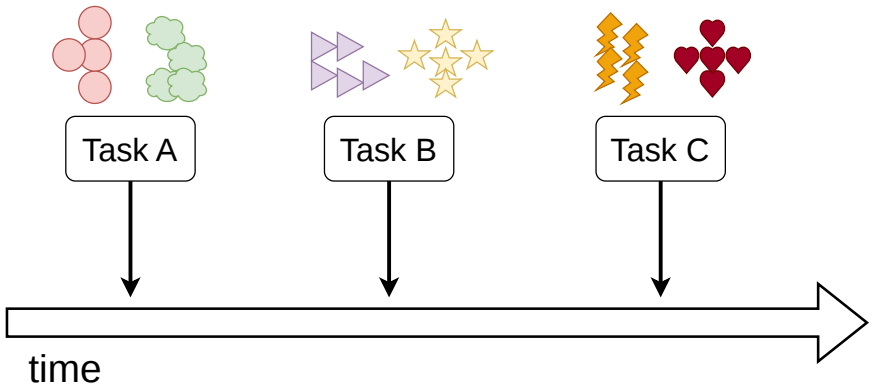


Figure 2.3: A sequential task arrival model for Continual Learning, depicting the transition from Task A to Task C over time, where each task represents a unique set of features or data distributions.

Without specific architectural or algorithmic safeguards, this sequential learning process inevitably triggers **catastrophic forgetting** [21]. This phenomenon occurs when the assimilation of new knowledge leads to the degradation or total loss of prior learning. As illustrated in Figure 2.4, the dilemma of forgetting becomes apparent when a *New Object Class* is introduced to an existing classifier, forcing a choice between two traditional but flawed strategies:

- **Fine-Tuning:** Updating the model using only the current task data is computationally efficient but leads directly to catastrophic forgetting [21]. Because the optimization process focuses solely on the new objective, the internal representations associated with previously learned classes are overwritten, causing the model to lose its original predictive capabilities.
- **Retraining from Scratch:** While joint training on the full historical dataset ($\mathcal{D}_{old} \cup \mathcal{D}_{new}$) technically prevents forgetting, it faces two significant hurdles. First is the issue of **Data Privacy and Availability**: in many real-world scenarios, the original data $\mathcal{D}_{old}$ is often deleted or inaccessible due to privacy regulations. Second is the **Computational Bottleneck**: as the system's lifespan increases, the cumulative dataset grows, making the energy and time required to retrain the model unsustainable in continuously evolving environments.

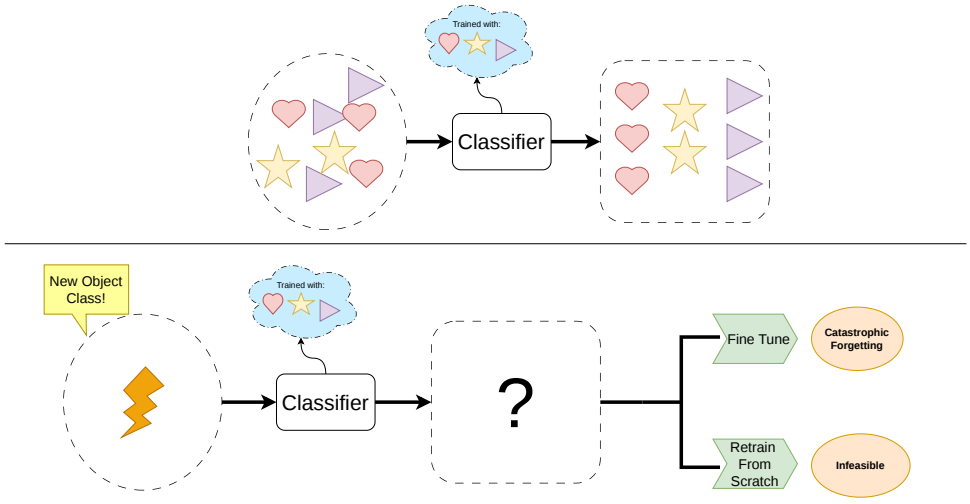


Figure 2.4: Standard Object Classifier comparison. The upper path shows standard inference, while the lower path illustrates the dilemma when a new class is introduced: the trade-off between catastrophic forgetting via fine-tuning and the infeasibility of retraining from scratch.

2.3.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. **Plasticity** refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. **Elasticity** (or stability), in contrast, emphasizes preserving the parameters and representations that are crucial for retaining old knowledge [7]. Achieving an appropriate equilibrium between these two properties is known as the **stability–plasticity dilemma** [21].

2.3.2 Main Paradigms

As shown in Figure 2.5, two principal paradigms define how continual learning problems are formulated. **Task-incremental learning (Task-IL)** assumes that the task identity is available during inference, allowing for task-specific components to be isolated [22]. **Class-incremental learning (Class-IL)**, on the other hand, does not provide the task identity at test time, requiring the model to distinguish between all classes seen across all tasks simultaneously [3].

2.3.3 Mitigation Strategies

To address the stability-plasticity dilemma, CL strategies are generally categorized into four main families: rehearsal-based, regularization-based, architecture-based, and representation-based (specifically prompt-based) methods.

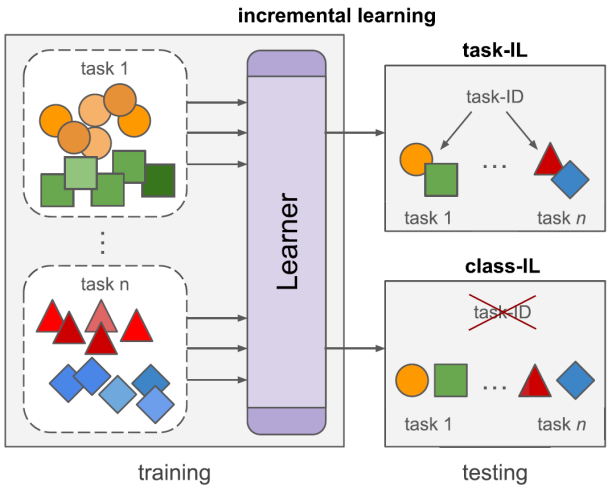


Figure 2.5: Class and Task Incremental Learning. Image taken from [3].

Rehearsal Methods

To preserve past performance, these methods reuse samples from previously learned tasks while training on the current task.

- **Experience Replay (ER)** [23]: Maintains a small episodic memory buffer containing a sample of old, representative examples to be interleaved with new data.
- **Average Gradient Episodic Memory (A-GEM)** [24]: Improves efficiency by using the buffer to constrain the direction of parameter updates, ensuring they do not interfere with past tasks.
- **Dark Experience Replay (DER)** [25]: Enhances ER by incorporating knowledge distillation on the logits of replayed examples to preserve the "dark knowledge" of the model.
- **Generative Replay** [26]: A non-data-storing method where a generator is trained to produce synthetic features or samples of previously learned classes, avoiding the privacy concerns of raw data storage.

Regularization Methods

These methods limit updates to parameters identified as important for prior learning by introducing penalty terms in the loss function.

- **Elastic Weight Consolidation (EWC)** [27]: Identifies critical parameters using the Fisher Information Matrix and restricts their movement during new task training.
- **Synaptic Intelligence (SI)** [28]: Calculates importance weights online by accumulating their contribution to the reduction of the loss.

- **Learning Without Forgetting (LwF)** [29]: A functional regularization method that uses knowledge distillation to ensure the model's outputs for new data remain consistent with the previous model's predictions.
- **Adaptive Plasticity Improvement (API)** [7]: Enforces an update direction orthogonal to the important subspaces of previous tasks using Dual Gradient Projection Memory (Dual-GPM).

Architecture-based Methods

These methods mitigate forgetting by modifying the network architecture to increase or reorganize its capacity.

- **Parameter Isolation**: Identifies and freezes parameters important for previous tasks –often through pruning or gradient masking– so only a subset of weights is updated for the current task [30].
- **Dynamic Expansion**: The architecture grows as new tasks arrive. For instance, [31] uses an expandable super-feature extractor followed by pruning, while [32] expands the network only when the current capacity is insufficient.

Representation and Prompt-based Methods

The emergence of foundation models (e.g., ViTs) has enabled parameter-efficient fine-tuning (PEFT) where the backbone remains frozen and learning is contained within small "prompts".

- **Learning to Prompt (L2P)** [33]: Uses a prompt pool where prompts are selected based on the similarity between input embeddings and learned keys.
- **Dual Prompt** [34]: Refines L2P by separating representations into fixed G-prompts (General) and task-specific E-prompts (Expert).
- **CODA-Prompt** [35]: Introduces a fully differentiable method where the prompt is a weighted sum of components dynamically calculated via attention mechanisms.
- **Advanced Prompting**: Recent methods like **InFLoRA** [38] and **DOLFIN** [36] apply low-rank adaptation and distributed learning, while **APT** [37] injects key-value tokens directly into transformer layers to reduce inference overhead.

2.3.4 Evaluation Metrics

To quantify performance, SOTA works [33] utilize two standard metrics derived from the test accuracy scores: the average accuracy and the average forgetting.

On the one hand, **Average Accuracy (A_t)**, presented on equation (2.15), measures overall performance after task t. It is the mean test accuracy across all tasks learned up to that point.

$$A_t = \frac{1}{t} \sum_{\tau=1}^{t} S_{t,\tau} \tag{2.15}$$

On the other hand, **Forgetting (F_t)**, presented on equation (2.16), quantifies how much the model has forgotten earlier tasks. It computes, for each past task, how much accuracy has dropped compared to its best historical score, and averages those drops.

$$F_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \left(\max_{k \leq t-1} S_{k,\tau} - S_{t,\tau} \right) \tag{2.16}$$

Chapter 3

Methodology

3.1 Proposed Architecture

As shown in 3.1, our CIL framework is built upon a simple yet structurally motivated decomposition: (i) a frozen feature extractor based on a visual foundation model, and (ii) a probabilistic classification layer that models each object category through an independent Gaussian distribution in the embedding space. Given an input image, the feature extractor maps it to a fixed high-dimensional representation. For each class, we maintain a class-conditional multivariate Gaussian characterized by its mean vector and covariance matrix, estimated incrementally from observed samples. Classification is performed according to the maximum likelihood principle, assigning the input to the class whose Gaussian model yields the highest likelihood under the stationary embedding representation. This modular separation between representation learning and statistical modeling enables incremental updates without backpropagation or parameter adaptation: adding a new class requires only estimating its distribution in the fixed feature space, while previously learned classes remain unaffected at the representation level.

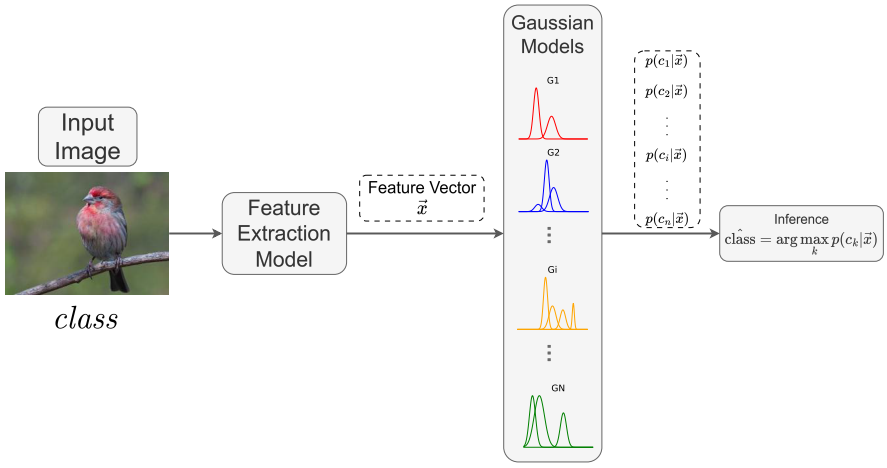


Figure 3.1: Block diagram of the proposed approach.

3.1.1 Datasets and Experimental Protocol

We follow the standard Continual Learning protocol where the model learns a sequence of disjoint tasks. The specific configurations for the datasets are as follows:

- Split CIFAR-100: This dataset is a widely utilized benchmark in the continual learning literature. It partitions the original CIFAR-100 [18] dataset into 10 disjoint tasks, with each task containing 10 distinct classes.
- Split ImageNet-R: This benchmark is constructed from ImageNet-R [17], a dataset containing renditions of ImageNet classes (e.g., art, cartoons, sketches). We randomly divide the 200 classes into 10 sequential tasks, with 20 classes per task.
- Split Caltech256:
- Split Joint Dataset:

3.2 Covariance Estimation

3.2.1 Curse of Dimensionality

3.3 Mixture of Gaussians

3.4 Hybrid Fusion and Score Normalization Strategies

In many classification pipelines, combining multiple metrics—such as the generative log-likelihood (Mahalanobis-based) and the discriminative angular similarity (Cosine-based)—can lead to superior performance.

3.4.1 Mahalanobis Distance and Temperature Scaling

The Mahalanobis distance is a scale-invariant metric that measures the distance between a point x and a distribution with mean μ and covariance Σ.

Distance and Likelihood Relation

The squared Mahalanobis distance D^2 is:

$$D^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

For a multivariate Gaussian distribution, the log-likelihood is:

$$\ln L(x) = -\frac{1}{2}\ln|\Sigma| - \frac{1}{2}D^2 - \frac{k}{2}\ln(2\pi) \tag{3.1}$$

This reveals that maximizing the log-likelihood is equivalent to minimizing the Mahalanobis distance (after accounting for the determinant of the covariance). This relationship is the basis for Mahalanobis-based classifiers, which are effective in high-dimensional embedding spaces where Euclidean distance fails to account for feature correlations.

Calibration via Temperature Scaling

Classifiers based on deep features often produce scores that are not well-calibrated probabilities. Temperature scaling is a post-processing technique that adjusts the log-likelihood scores before they are passed through a softmax function.The score for class i is modified by a scalar $T > 0$:

$$\text{Score}_i = \frac{1}{T}\left(\ln A_i - \frac{1}{2}D_i^2\right) \tag{3.2}$$

where $\ln A_i$ is the log-normalization constant. An optimal T is typically learned using a validation set to minimize negative log-likelihood (NLL). $T > 1$ "softens" the distribution (reducing overconfidence), while $T < 1$ "sharpens" it. This calibration is critical for tasks requiring reliable uncertainty estimation, such as out-of-distribution detection or safety-critical classification.

3.5 Backbones comparison

Chapter 4

Results

4.1 Covariances Estimation and MoGs

Table 4.1: Results. A_{10} and F_{10} represent average accuracy and forgetting for the 10-task setting. Our proposed methods are highlighted in grey.

Method	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech256**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
L2P	82.76 ± 1.17	7.86 ± 0.39	63.49 ± 0.40	6.85 ± 0.42	—	—
DualPrompt	85.07 ± 0.49	5.57 ± 0.20	68.50 ± 0.52	5.14 ± 0.18	—	—
DualPrompt-PGP	86.92 ± 0.05	5.35 ± 0.19	69.34 ± 0.05	4.53 ± 0.04	—	—
CODA-Prompt	87.00 ± 0.38	4.78 ± 0.24	74.24 ± 0.56	4.92 ± 0.21	—	—
CPrompt	87.83 ± 0.37	4.88 ± 0.79	76.71 ± 0.61	4.66 ± 0.93	—	—
ConvPrompt	88.87 ± 0.33	4.75 ± 0.15	77.86 ± 0.25	4.33 ± 0.24	—	—
RainbowPrompt	89.86 ± 0.11	3.44 ± 0.26	79.09 ± 0.13	3.90 ± 0.23	—	—
APT	88.88 ± 0.65	3.47 ± 0.55	79.40 ± 0.47	4.38 ± 0.46	—	—
Diag-Cov (ours)	86.07 ± 0.00	4.30 ± 0.12	77.12 ± 0.00	7.44 ± 0.17	90.06 ± 0.00	1.69 ± 0.36
Full-Cov (ours)	88.48 ± 0.00	4.18 ± 0.00	75.08 ± 0.00	5.07 ± 0.03	90.01 ± 0.01	2.16 ± 0.05
Hybrid-Cov (ours)	88.49 ± 0.00	4.18 ± 0.00	75.12 ± 0.00	5.03 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
Shared-Diag-Cov (ours)	86.54 ± 0.00	4.61 ± 0.10	75.12 ± 0.01	4.80 ± 0.24	93.00 ± 0.00	2.00 ± 0.11
Shared-Full-Cov (ours)	88.56 ± 0.00	3.76 ± 0.11	77.06 ± 0.01	3.97 ± 0.30	93.54 ± 0.00	2.05 ± 0.29
MOG2-D (ours)	86.96 ± 0.00	4.24 ± 0.02	70.15 ± 0.00	4.93 ± 0.02	91.41 ± 0.00	1.67 ± 0.30
MOG2-F (ours)	85.01 ± 0.00	5.13 ± 5.13	73.42 ± 0.00	5.49 ± 0.26	93.95 ± 0.00	2.14 ± 0.11
MOG2-H (ours)	84.68 ± 0.00	5.26 ± 0.22	73.52 ± 0.00	5.47 ± 0.23	93.93 ± 0.00	2.13 ± 0.10
MOG4-D (ours)	87.80 ± 0.00	4.02 ± 0.06	72.52 ± 0.01	5.06 ± 0.15	91.12 ± 0.00	1.63 ± 0.31
MOG4-F (ours)	88.87 ± 0.00	3.99 ± 0.00	75.07 ± 0.00	5.07 ± 0.05	94.39 ± 0.00	2.12 ± 0.08
MOG4-H (ours)	88.88 ± 0.00	4.04 ± 0.00	75.01 ± 0.00	5.12 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
OAS (ours)	84.38 ± 0.00	3.90 ± 0.07	71.47 ± 0.00	4.57 ± 0.29	86.70 ± 0.00	1.90 ± 1.77
LW-I (ours)	84.78 ± 0.00	3.93 ± 0.09	71.91 ± 0.00	4.54 ± 0.43	89.89 ± 0.00	1.50 ± 0.08
LW-D (ours)	82.85 ± 0.00	3.94 ± 0.28	70.35 ± 0.00	4.62 ± 0.95	90.04 ± 0.00	1.66 ± 0.30

4.2 Hybrid Fusion and Score Normalization Strategies

4.3 Backbones comparison

Table 4.2: Performance results on Split Joint Datasets.

Method	**Split Joint Datasets**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
Diag-Cov		
Full-Cov		
Hybrid-Cov		
Shared-Full-Cov		
Shared-Diag-Cov		
MOG2-D		
MOG2-F		
MOG2-H		
MOG4-D		
MOG4-F		
MOG4-H		
OAS		
LW-I		
LW-D		

Table 4.3: Detailed performance results across multiple backbones and datasets.

Method	**Backbone**	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech256**		**Split Joint**	
		$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
Diag-Cov	DINOv2 ViT-m ViT-b								
Full-Cov	DINOv2 ViT-m ViT-b								

Chapter 5

Discusion

Chapter 6

Conclusion

6.1 Future Work

At this stage, the integration of Open World Discovery appears to be a straightforward extension of the existing framework. Since our approach provides a certainty score, implementing an Out-of-Distribution (OOD) threshold allows the system to flag unknown classes with minimal architectural changes. This addresses the objective of moving beyond a closed-set classifier toward an adaptive, "discovery-ready" system.

In contrast, the transition to Real-Time Object Detection requires a more strategic approach. While previous experiments utilizing a Region Proposal Network (RPN) XXXAnnexXXX yielded high localization accuracy, the computational overhead of a dual-model pipeline (RPN + Feature Extractor) compromises the inference speed. By leveraging the spatial tokens already present in the DINOv2 backbone, we can identify regions of interest using simple mathematical decomposition. This ensures that the system remains lightweight and consistent with the unsupervised nature of the feature extractor.

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